

Skill & Product Library / Compliance

Automation Workflow

Live in production

EOR Compliance Checker

AI deal screening on the live sales pipeline: every deal gets an auditable compliance verdict the moment it enters.



100+ deals wrongly stuck in review correctly cleared; zero production write errors

n8n

Azure OpenAI

HubSpot

Avoma

compliance

rules engine

Overview

An automated compliance-screening system that evaluates every deal entering a global HR-tech (EOR) platform's sales pipeline against the company's official 30-sector compliance worksheet, and writes an auditable verdict back to the CRM. An LLM handles only the fuzzy work (mapping messy CRM data to canonical sectors); a unit-tested JavaScript rules engine makes every tier decision deterministically. Result: all open deals in the pipeline screened, over a hundred deals wrongly stuck in "review required" correctly cleared while ineligible-deal detection more than doubled — every verdict traceable to a named rule, with a perfect production write record.

The Problem

Deals reached contracting and KYC before anyone checked whether the role or industry was even eligible for Employer-of-Record services — compliance found blockers at signature time, killing deals late, wasting effort, and eroding trust. Screening was ad-hoc and reactive.

How It Works

1. A CRM deal workflow fires on pipeline entry and posts the deal ID to an **n8n** webhook — n8n is a workflow-automation platform running the screening pipeline (46 nodes).

2. n8n re-fetches everything authoritative by ID via the **CRM API**: deal, contacts (job titles), notes, company — plus call transcripts from a conversation-intelligence tool that records sales calls.
3. An **enterprise-hosted LLM** does *semantic mapping only*: reading messy CRM values ("Computer Software" ≈ "Software Development") and pass-through detection (a staffing firm placing oil & gas workers is screened as oil & gas).
4. A **deterministic JavaScript matrix engine** — the 30-sector worksheet transcribed into unit-tested code — applies three independent judgments: viability tier (1/2/3), preliminary flags (executive role / legal-entity conflict), and hard screens (licenses, union-CBA, security clearance) that force Tier 3. Confidence is a report field; it never changes the tier.
5. The verdict is written to five CRM deal properties (tier, eligibility flag, preliminary flag, screen-required flag, notes with cited evidence), and the deal owner gets an email notification if Tier 2/3.
6. An hourly n8n safety-net sweep screens any open deal still missing a tier — a zero-blank guarantee even if a real-time event is missed.



Stack & Integrations

Tool	Role	Access needed
n8n cloud	Orchestration — webhook, batch, dry-run, and hourly-sweep workflows	Workspace + workflow import
Enterprise LLM (GPT-4 class)	Semantic sector mapping + note prose only	Deployment + API credential
Custom JS rules engine	Deterministic tiers, gates, alias resolution	Ships inside the workflow
CRM API (HubSpot)	Deal/contact/note/company reads; property writes	Private-app token: deals + contacts read-write
Call-intelligence API	Call transcripts & summaries	API key, read-only
Email (Outlook)	Owner notifications on Tier 2/3	OAuth (mail send)

Setup Process

1. Create the five compliance deal properties in your CRM.
2. Import the three workflow JSONs (webhook, batch updater, report-only tester) into n8n; bind CRM, LLM, call-intelligence, and email credentials.

3. Build the CRM deal workflow: enroll on target stages, "Send a webhook" action posting the deal ID to the n8n URL.
4. Run the **report-only dry-run harness** against all open deals — zero CRM writes — and validate the tier distribution.
5. Run the batch updater to screen the full pipeline; verify sample writes per tier.
6. Enable the production webhook + hourly sweep; park the previous version off (not deleted) as instant rollback.

What You Need

- CRM private-app token (deals/contacts/notes read-write); n8n cloud workspace
- Enterprise LLM deployment; call-intelligence API key; email/mail credential
- The official sector-by-sector compliance worksheet (source of the rules) · **Time-to-deploy: ~1 week** including dry-run validation



Results

- **Live in production**; hundreds of open deals screened across the full pipeline
- **Over 100 deals wrongly stuck in "review required" correctly cleared**; ineligible-deal detection more than doubled, all traceable to a named worksheet trigger
- **Perfect production write record** (zero errors); full unit-test coverage; well over 100 correct tier-downgrade regression moves; final portfolio reached zero "Insufficient Data" verdicts after an alias-resolution upgrade
- Live end-to-end validation on a real deal: dozens of nodes, zero errors

Challenges & Fixes

- Most "Insufficient Data" verdicts → root cause: contacts were never fetched in earlier versions → contact-fetch chain added
- Over-conservative tiers — thin data demoted clean sectors → re-architecture: confidence reports but never changes the tier
- A variable-reassignment bug surfaced on the first live production test → caught by the verification step, fixed same day, re-test passed
- Several analytics properties didn't exist in the CRM and silently failed writes → folded into the notes field

Demonstrates

LLM + deterministic hybrid design · regulated-domain AI with full audit trails · zero-risk deployment methodology (dry-run harness, rollback parking) · multi-API orchestration.

STACK & INTEGRATIONS

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WHAT IT NEEDS

- ✓ CRM private-app token (deals/contacts/notes read-write)
- ✓ n8n cloud workspace
- ✓ Enterprise LLM deployment (e.g. Azure OpenAI GPT-4)
- ✓ Call-intelligence API key (e.g. Avoma)
- ✓ Email/mail OAuth credential
- ✓ A written sector-by-sector compliance worksheet



DISCUSS THIS BUILD

Happy to walk through the architecture, the failures along the way, or how this could translate to your team.

Email Patrick

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